

What Observable Properties Are Most Determinative of Dark Matter Fractions in Spiral Galaxies Like the Milky Way, and can we use those properties to make estimations of the Dark Halo Mass?

Aarav Sharma, Samarth Singhal, Arnav Saraf

Historical Justification

Astronomers have studied the motions of stars and galaxies for more than a century. Early models assumed that the gravity from visible matter could explain how galaxies move. During the 1930s, Fritz Zwicky found that galaxies in the Coma Cluster were moving too quickly to be held together by visible matter alone. Decades later, Vera Rubin and Kent Ford measured rotation curves of spiral galaxies and found that stars far from galactic centers continued moving at unexpectedly high speeds. These observations became some of the strongest evidence that galaxies contain large amounts of unseen matter called dark matter.

Since then, observations from rotation curves, gravitational lensing, galaxy clusters, and the cosmic microwave background have consistently supported the existence of dark matter. Although scientists estimate that dark matter makes up most of the matter in the universe, its physical nature remains unknown. One important open question is why some spiral galaxies appear to have larger dark matter fractions than others and which observable properties are most closely connected to those differences.

Modern sky surveys have produced large, carefully calibrated datasets containing measurements such as stellar mass, gas mass, luminosity, radius, metallicity, rotation velocity, and star formation rate. These datasets enable the search for potentially hidden but significant relationships that cannot be easily discerned by conventional statistical techniques. Understanding these relationships could lead to advances in our models of galaxy formation and evolution and enable astronomers to make better decisions about how to allocate future observations.

Our project extends decades of research done by astronomers around the world. Scientific progress is made by building on past achievements while asking new questions through novel approaches. Isaac Newton famously articulated this principle by observing that researchers stand on the shoulders of giants. Our project follows this tradition by using the reliable observational data amassed through past research and by applying modern machine learning techniques to answer an important question in galactic astrophysics.

Reliable Sources Summary

1. SPARC Catalog by Lelli, McGaugh, and Schombert. Introduces a high quality catalog of nearby spiral galaxies with rotation curves and photometry that is widely used for dark matter studies.
2. SPARC Database. Provides stellar masses, gas masses, luminosities, rotation curves, and other measurements suitable for statistical analysis.
3. Vera Rubin's work on galaxy rotation curves established observational evidence that visible matter alone cannot explain galactic rotation speeds.
4. Planck Collaboration cosmology results summarize the current understanding of the matter and energy content of the universe and provide strong evidence for dark matter.
5. Sloan Digital Sky Survey has created one of the largest astronomical datasets available and has enabled thousands of studies on galaxy properties and evolution.

References

Lelli, F., McGaugh, S. S., & Schombert, J. M. (2016). The SPARC database. *The Astronomical Journal*, 152(6), 157.

<https://iopscience.iop.org/article/10.3847/0004-6256/152/6/157>

SPARC Database. <https://astroweb.cwru.edu/SPARC/>

SPARC paper PDF. <https://astroweb.case.edu/ssm/papers/AJv152n157.pdf>

Rubin, V. C., Ford, W. K., & Thonnard, N. (1980). Rotational properties of spiral galaxies.

Planck Collaboration. (2020). Planck 2018 results. VI. Cosmological parameters.

Sloan Digital Sky Survey. <https://www.sdss.org/>

Zwicky, F. (1933). Die Rotverschiebung von extragalaktischen Nebeln